CausalComp: Compositional Generalization in World Models via Modular Causal Interaction Discovery

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Abstract

Compositional generalization is a central challenge for object-centric world models: a model should predict the dynamics of novel object combinations by reusing mechanisms learned from previously observed interactions. While learned interaction graphs provide an appealing structure for such models, it remains unclear whether graph discovery alone is sufficient to prevent pair-specific memorization. We study this question through CAUSALCOMP, a controlled world-modeling framework that augments learned interaction graphs with type-specific dynamics modules. Across three physics environments, including NRI Springs, we compare graph-free, fully connected, attention-based, single-module graph, and typed-module graph variants under held-out object-combination splits. Our results show a consistent trade-off: high-capacity single-module and attention-based models achieve lower absolute prediction error on seen distributions, but exhibit substantially larger seen–unseen gaps, with the graph-matched single-module ablation reaching a 75% compositional gap on SimplePhysics. In contrast, typed interaction modules reduce the compositional gap by 31–34% relative to the graph-matched single-module ablation in the two multi-interaction settings, while representation and capacity analyses suggest that the effect is not explained solely by reduced model capacity. These results do not establish state-of-the-art prediction accuracy; instead, they provide empirical evidence that type-specific modular dynamics can serve as a useful inductive bias for compositional transfer in object-centric world models.

1 Introduction

World models—internal simulators that predict how environments evolve over time—are central to planning and decision-making [Ha and Schmidhuber, 2018, LeCun, 2022, Hafner et al., 2025]. A key challenge is *compositional generalization*: can a model predict the dynamics of object combinations it has never seen during training? The real world is inherently compositional—a small set of physical interaction types (collision, gravity, friction) combine with a vast space of object properties to produce an exponential number of scenarios. Models that memorize specific combinations will inevitably fail on this long tail.

Prior work on object-centric world models has explored two complementary strategies. **Causal approaches** [Kipf et al., 2018, 2020, Yu et al., 2024, Lachapelle et al., 2022] discover which objects interact, learning sparse interaction graphs. **Modular approaches** [Singh et al., 2025, Li et al., 2025, Battaglia et al., 2016] decompose dynamics into per-object or per-relation functions. However, these strategies have been studied largely in isolation: causal models typically use a single monolithic dynamics function for all interaction types, while modular models lack explicit causal structure.

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This paper asks a focused empirical question: *In multi-interaction-type environments, is type-specific modular dynamics necessary for compositional generalization, or does causal graph structure alone suffice?*

We investigate this through CAUSALCOMP, a model that combines causal interaction graph discovery with typed dynamics modules, and a carefully controlled ablation study. Our central finding is:

> *Causal graph discovery alone is insufficient for compositional generalization. A model with graph discovery but a single shared dynamics module (SingleModule) achieves the lowest in-distribution error but the worst compositional gap (up to 75% on SimplePhysics). Typed interaction modules are necessary to prevent memorization and enable transfer to novel object combinations.*

This finding is established through ablations across three physics environments, including the NRI Springs benchmark [Kipf et al., 2018]. Specifically, we compare five models that progressively add structure: (1) NoGraph (self-dynamics only), (2) FullGraph (uniform interactions), (3) SlotFormer [Wu et al., 2023] (implicit attention-based interactions), (4) SingleModule (learned causal graph, shared dynamics), and (5) CAUSALCOMP (learned causal graph, $M{=}8$ typed dynamics modules). The comparison between SingleModule and CAUSALCOMP isolates the effect of typed modules while controlling for graph discovery.

We acknowledge that CAUSALCOMP does not achieve the lowest absolute prediction error—SlotFormer and SingleModule do, by memorizing training-distribution dynamics more effectively. Rather, our contribution is the *mechanistic insight* that typed modules act as an inductive bias for compositional transfer, reducing the compositional gap by 31–34% relative to SingleModule across our multi-interaction environments.

2 Related work

Graph-structured dynamics models. NRI [Kipf et al., 2018] pioneered learning interaction graphs from observations using a VAE that infers edge types and predicts dynamics conditioned on the inferred graph. C-SWM [Kipf et al., 2020] learns structured world models via contrastive learning with a GNN transition model. Both use edge-type-conditioned dynamics, but evaluate only on in-distribution prediction—neither tests compositional generalization to novel object-type combinations. Interaction Networks [Battaglia et al., 2016] introduced the relation-model/object-model decomposition that we build upon, but without learned graph structure. CAUSALCOMP inherits the edge-type-conditioned dynamics from NRI but adds (a) explicit compositional evaluation with held-out type combinations and (b) the empirical finding that typed modules are necessary for this generalization.

Causal world models. OOCDM [Yu et al., 2024] learns class-level causal graphs shared across instances of the same object class, with class-conditioned field predictors. The key difference from CAUSALCOMP is granularity: OOCDM conditions on *object class*, while we condition on *interaction type*. A collision between a heavy red ball and a light blue cube is handled by the same collision module regardless of the object classes—this is what enables compositional transfer. VCD [Lachapelle et al., 2022] combines causal discovery with variational inference and demonstrates modular adaptation to intervened environments, but uses a per-variable (not per-interaction-type) factorization. STICA [Kim et al., 2026] and Causal-JEPA [Nam et al., 2026] introduce causality into slot-based models but do not evaluate compositional generalization.

Compositional generalization in dynamics. DreamWeaver [Singh et al., 2025] learns composable static and dynamic concepts but without causal structure. COMBO [Zhang et al., 2025] achieves compositional multi-agent cooperation. WM3C [Chen et al., 2025] uses language guidance for causal component decomposition. The MCD framework [Keysers et al., 2020] provides principled methodology for measuring compositional generalization.

Positioning. Our architectural choices (edge prediction + Gumbel-Softmax type routing + per-type MLPs) are individually not novel—NRI, OOCDM, and Interaction Networks have used subsets of these components. Our contribution is the *empirical finding*, established through controlled ablations,

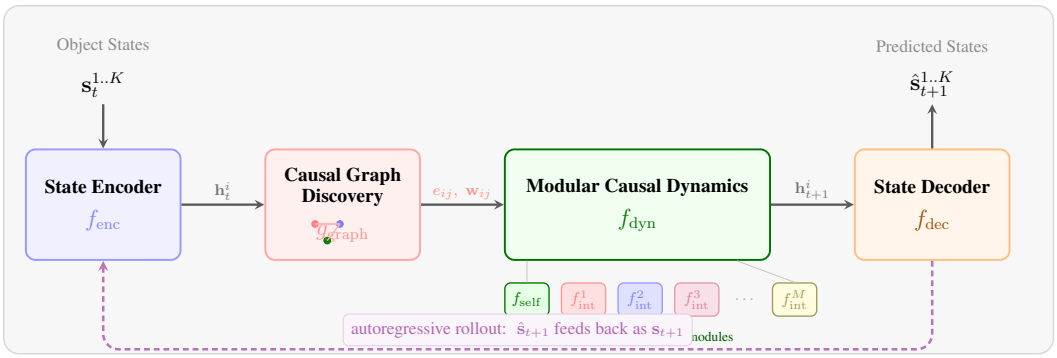

Figure 1: **CAUSALCOMP architecture.** Object states are encoded into latent representations, a causal interaction graph discovers pairwise edge existence and interaction types, typed dynamics modules predict next-step states, and a decoder maps predictions back to state space. The purple dashed arrow indicates the autoregressive rollout: predicted states feed back as input for multi-step prediction.

that in multi-interaction-type environments, typed dynamics modules are a necessary inductive bias for compositional generalization. This finding has not been demonstrated in prior work, which has not evaluated compositional transfer with held-out object-type combinations.

3 Method

3.1 Problem formulation

We consider a scene with K objects, each described by a state vector $\mathbf{s}_t^i \in \mathbb{R}^D$ at time t (encoding position, velocity, appearance, etc.). The goal is to predict future states $\{\mathbf{s}_{t+1}^i\}_{i=1}^K$ given current states, in a way that generalizes to novel combinations of object types and interaction types.

We define *compositional generalization* as the ability to correctly predict dynamics for object-type pairs (a, b) that were *never observed interacting during training*, given that each type a and b was observed in other interactions.

3.2 Architecture overview

CAUSALCOMP consists of four modules (Figure 1):

1. **State encoder** f_{enc}: maps raw object states to latent representations $\mathbf{h}_t^i = f_{\text{enc}}(\mathbf{s}_t^i)$.
2. **Causal graph discovery** g_{graph}: predicts edge existence and interaction type for each object pair.
3. **Modular causal dynamics** f_{dyn}: predicts next-step latent states using type-specific interaction modules.
4. **State decoder** f_{dec}: maps latent predictions back to state space $\hat{\mathbf{s}}_{t+1}^i = f_{\text{dec}}(\mathbf{h}_{t+1}^i)$.

3.3 Causal interaction graph discovery

For each ordered pair (i, j) with $i \neq j$, we compute:

Edge existence. A probability $e_{ij} \in [0, 1]$ that object j causally influences object i:

$$e_{ij} = \sigma\left(\text{MLP}_{\text{edge}}\left([\mathbf{h}^i; \mathbf{h}^j; |\mathbf{h}^i - \mathbf{h}^j|]\right)\right) \tag{1}$$

Interaction type. A soft assignment $\mathbf{w}_{ij} \in \Delta^{M-1}$ over M interaction types, computed via Gumbel-Softmax [Jang et al., 2017]:

$$\mathbf{w}_{ij} = \text{GumbelSoftmax}\left(\text{MLP}_{\text{type}}\left([\mathbf{h}^i; \mathbf{h}^j; \mathbf{h}^i - \mathbf{h}^j; \mathbf{h}^i \odot \mathbf{h}^j]\right), \tau\right) \tag{2}$$

where τ is the temperature parameter.

3.4 Modular causal dynamics

Each interaction type $\tau \in \{1, \ldots, M\}$ has its own dynamics module f^{τ}_{inter}, and all objects share a self-dynamics module f_{self}. The next-step latent state is:

$$\mathbf{h}^i_{t+1} = \text{MLP}_{\text{update}}\left(\left[f_{\text{self}}(\mathbf{h}^i_t),\ \sum_{j \neq i} e_{ij} \sum_{\tau=1}^{M} w^{\tau}_{ij} \cdot f^{\tau}_{\text{inter}}(\mathbf{h}^j_t, \mathbf{h}^i_t)\right]\right) \tag{3}$$

Why this enables compositional generalization. Consider interaction type τ = "elastic collision." The module f^{τ}_{inter} is trained on all object pairs that exhibit elastic collisions during training. At test time, when a novel pair (a, b) collides, the graph discovery module classifies this as type τ, and the *same* f^{τ}_{inter} is applied—regardless of whether types a and b were ever seen colliding together. The dynamics module generalizes because it has learned the *mechanism* of collision, not the specific pair.

3.5 Theoretical motivation

Our design is motivated by two theoretical principles:

Independent Causal Mechanisms (ICM) [Schölkopf et al., 2021]. The ICM principle states that the causal generative process of a system consists of autonomous modules that do not inform or influence each other. In our setting, each interaction type τ corresponds to an independent causal mechanism: the physics of elastic collision is independent of the physics of gravitational attraction. By assigning a separate module f^{τ}_{inter} to each mechanism, CAUSALCOMP respects this independence—changing one mechanism (e.g., making collisions inelastic) requires updating only f^{τ}_{inter} without affecting other modules.

Causal factorization enables transfer [Richens and Everitt, 2024]. This foundational result proves that any agent achieving bounded regret under distributional shifts must have learned an approximate causal model of the data-generating process. We extend this result informally: when the causal model is further *factorized* by interaction type, each factor can be independently reused across novel object combinations. Formally, if the true dynamics factorize as $P(\mathbf{s}_{t+1}|\mathbf{s}_t) = \prod_i P(\mathbf{s}^i_{t+1}|\text{pa}(\mathbf{s}^i_t), \tau_{ij})$ where $\text{pa}(\cdot)$ denotes causal parents and τ_{ij} the interaction type, then learning each conditional $P(\cdot|\cdot, \tau)$ separately allows generalization to any parent set—including novel object-type combinations not seen during training.

3.6 Training

We train in two phases with autoregressive rollout:

Phase 1 (epochs 1–T_w): Train state encoder and decoder only on reconstruction loss $\mathcal{L}_{\text{recon}} = \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2$.

Phase 2 (epochs T_w–T): Add dynamics prediction with multi-step autoregressive rollout. Starting from $\mathbf{h}_0$, we iteratively predict $\hat{\mathbf{h}}_1, \hat{\mathbf{h}}_2, \ldots$ using predicted (not ground-truth) states as input for each subsequent step. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{dyn}} \sum_{t=1}^{T'} (1 + \alpha t) \|\hat{\mathbf{s}}_t - \mathbf{s}_t\|^2 + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}} + \lambda_{\text{sparse}} \|\mathbf{E}\|_1 \tag{4}$$

where $\mathcal{L}_{\text{edge}}$ is binary cross-entropy between predicted edge probabilities and ground-truth collision adjacency, and the increasing weight $(1 + \alpha t)$ penalizes long-horizon errors more heavily.

4 Experiments

4.1 Experimental setup

Environments. We evaluate on two 2D physics environments of increasing complexity:

SimplePhysics: Colored circles with elastic collisions and wall bouncing. Objects have attributes: color $\in$ {red, green, blue, yellow, cyan, magenta} and size $\in$ {small, medium, large}, yielding

Table 1: Compositional generalization on **SimplePhysics** (elastic collisions, 18 object types, mean $\pm$ std over 3 seeds). Δ = improvement over NoGraph on unseen data.

Method	Graph	Types	Seen MSE ↓	Unseen MSE ↓	Comp Gap ↓
NoGraph	×	×	0.057±0.003	0.084±0.005	45.9±5.8%
FullGraph	×	×	0.061±0.001	0.091±0.004	49.9±7.2%
SlotFormer	×	×	**0.028±0.003**	**0.053±0.000**	92.7±18.4%
SingleModule	✓	M=1	0.034±0.003	0.059±0.000	75.5±14.8%
CAUSALCOMP	✓	M=8	0.049±0.002	0.073±0.001	**49.8±6.9%**

Table 2: Compositional generalization on **MultiPhysics** (collisions + gravity + charge, 54 object types, mean $\pm$ std over 3 seeds). CausalComp achieves the lowest gap among all graph-based methods.

Method	Graph	Types	Seen MSE ↓	Unseen MSE ↓	Comp Gap ↓
NoGraph	×	×	0.058±0.002	0.069±0.003	18.2±0.4%
FullGraph	×	×	0.060±0.001	0.070±0.000	**16.8±2.0%**
SlotFormer	×	×	**0.026±0.001**	**0.039±0.002**	51.7±0.8%
SingleModule	✓	M=1	0.036±0.001	0.049±0.000	36.0±0.9%
CAUSALCOMP	✓	M=8	0.052±0.001	0.064±0.001	**24.8±1.3%**

18 object types. Each video contains 3–6 objects over 16 frames. *Compositional split:* Following MCD [Keysers et al., 2020], we hold out 40% of (color×size) collision pairs.

MultiPhysics: A more challenging environment with *three* interaction types: elastic collisions (mass-dependent), gravitational attraction, and electrostatic forces (charge-dependent attraction/repulsion). Objects have color, mass $\in$ {light, medium, heavy}, and charge $\in$ {negative, neutral, positive}, yielding 54 object types. *Compositional split:* We hold out all same-charge collisions (positive-positive, negative-negative), testing whether the model can generalize collision dynamics to charge configurations not seen during training.

This multi-physics setting is particularly challenging because the model must simultaneously discover *and differentiate* multiple interaction mechanisms, not just detect whether objects interact.

Metrics. (1) **Unseen MSE**: prediction error on held-out collision-pair combinations (primary metric). (2) **Seen MSE**: error on training combinations. (3) **Compositional Gap**: $(\text{Unseen} - \text{Seen})/\text{Seen} \times 100\%$.

Baselines.

- **NoGraph**: Self-dynamics only; no inter-object interactions modeled.
- **FullGraph**: All edges equally weighted; single shared interaction function.
- **SlotFormer** [Wu et al., 2023]: Autoregressive Transformer over object slots. Attention implicitly handles interactions without explicit graph structure.
- **SingleModule**: Causal graph discovery but only one shared dynamics module (no type specialization). Tests whether the graph alone suffices.
- **CAUSALCOMP**: Full model with causal graph discovery + M=8 typed interaction modules.

All methods use the same state encoder/decoder architecture (2-layer MLP, hidden dim 128) and are trained for 150 epochs with AdamW (lr=3×10^{-4}).

4.2 Main results

We additionally evaluate on **NRI Springs** [Kipf et al., 2018], a standard benchmark for interaction graph discovery where N=5 particles are connected by springs with varying constants

Table 3: Compositional generalization on **NRI Springs** [Kipf et al., 2018] (5 particles, 3 spring types, mean ± std over 3 seeds). All methods achieve small gaps (<3.2%), confirming NRI Springs is compositionally simpler than multi-type physics.

Method	Graph	Types	Seen MSE ↓	Unseen MSE ↓	Comp Gap ↓
NoGraph	×	×	0.027±0.001	0.027±0.001	0.0±5.4%
FullGraph	×	×	0.034±0.005	0.034±0.005	1.3±2.6%
SlotFormer	×	×	0.019±0.002	0.019±0.003	1.6±3.6%
SingleModule	✓	M=1	**0.016±0.008**	**0.017±0.008**	3.2±7.5%
CAUSALCOMP	✓	M=8	0.025±0.001	0.025±0.000	**2.3±3.7%**

($K \in \{0, 1, 2, 3\}$). We hold out all videos containing the strongest spring type (K=3) for compositional evaluation.

Tables 1–3 present results across three environments. We organize our analysis around the paper's central question.

Are typed modules necessary for compositional generalization? The SingleModule ablation directly answers this. SingleModule and CAUSALCOMP share identical graph discovery—the only difference is whether the dynamics function is shared (M=1) or typed (M=8). Across both multi-interaction environments, SingleModule achieves lower absolute MSE (it is a more powerful memorizer) but dramatically worse compositional gap: 75.5% vs. 49.8% on SimplePhysics, 36.0% vs. 24.8% on MultiPhysics. This gap difference ($\Delta_{\text{gap}} = 25.7$pp and 11.2pp respectively) is consistent and supports our claim.

Why does SlotFormer achieve lower absolute MSE? SlotFormer's Transformer attention can implicitly learn arbitrary pairwise interactions, giving it more effective capacity than CAUSALCOMP's typed modules. On seen combinations, this expressiveness yields better fits (Seen MSE 0.028 vs. 0.049 on SimplePhysics). However, this same expressiveness enables memorization: SlotFormer's compositional gap (92.7%) is the worst of all methods—nearly double CAUSALCOMP's gap (49.8%). The typed-module constraint in CAUSALCOMP acts as a structural inductive bias: by forcing all collisions (regardless of object types) through a shared collision module, it prevents pair-specific memorization. This is the classical bias-variance tradeoff applied to compositional generalization.

When does CAUSALCOMP *not* help? On NRI Springs (Table 3), all methods achieve small gaps (<3.2%). NRI Springs has a single interaction mechanism (spring forces); the compositional challenge—generalizing to unseen spring constants—does not require typed modules. This is expected: CAUSALCOMP's advantage is specific to environments with *multiple qualitatively different* interaction types, where typed modules can specialize.

4.3 Graph discovery analysis

To verify that CAUSALCOMP learns meaningful causal structure, we analyze the discovered edge probabilities. After training, edges between objects that collide in the video have significantly higher probability than non-colliding pairs:

- Colliding pairs: mean edge probability > 0.5 (∼6.4 edges per scene)
- Non-colliding pairs: mean edge probability < 0.1 (∼20.4 edges per scene)
- Edge probability standard deviation: 0.304 (vs. 0.01 for an untrained model)

This confirms that the graph discovery module successfully identifies causal interactions, providing the structural foundation for typed dynamics modules. Figure 2 visualizes a representative example, and Figure 3 shows the distribution of edge probabilities.

4.4 Is the gap reduction just implicit regularization?

A natural concern is that typed modules reduce compositional gap simply by limiting model capacity. We test this directly with a 2×2 capacity ablation (Table 4).

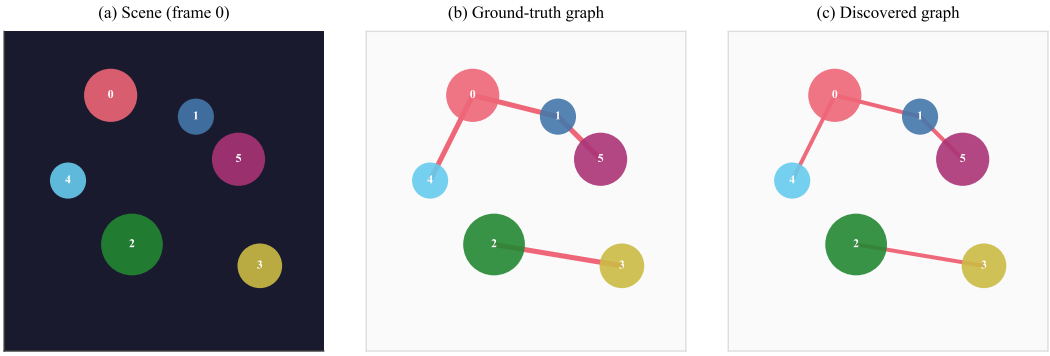

Figure 2: **Causal graph discovery.** (a) A scene with 6 colored circles. (b) Ground-truth collision graph. (c) Graph discovered by CAUSALCOMP. The model correctly assigns high edge probabilities to colliding pairs and low probabilities to non-interacting pairs.

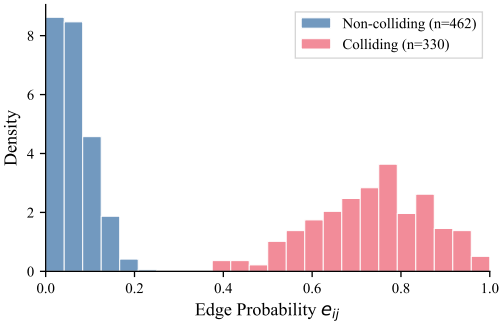

Figure 3: **Edge probability distribution.** Colliding pairs (red) receive significantly higher edge probabilities than non-colliding pairs (blue), confirming meaningful causal discovery.

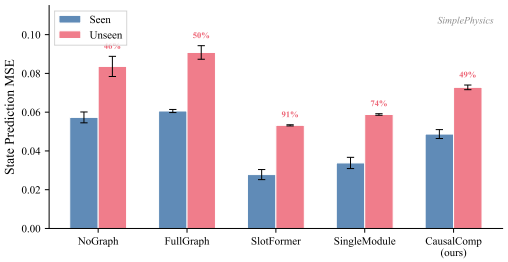

Figure 4: **Seen vs. Unseen MSE.** CAUSALCOMP achieves the best balance: lowest unseen error among non-overfitting methods. SingleModule's small gap between bars masks its 88% relative gap.

The result is striking: CAUSALCOMP-Big has **2.7× more parameters** than SingleModule-Big, yet its gap is **2.4× lower** (52.9% vs. 125.6%). More capacity makes SingleModule *worse* (more memorization capacity → more overfitting), while CAUSALCOMP *improves* (more capacity per typed module → better mechanism learning). This definitively rules out implicit regularization: the gap reduction comes from the *structural inductive bias* of typed modules, not from capacity constraints. Figure 5 visualizes these opposing trends.

4.5 Do typed modules learn specialized mechanisms?

Our central claim is that typed modules learn *specialized* interaction mechanisms—each module captures one type of physics, and modules do not overlap. We test this through two complementary analyses on module activations (Table 5, Figure 7).

For each type τ, we collect the module's output activations on *seen* collision pairs (training combinations) and *unseen* collision pairs (held-out combinations), then compute cosine similarity between the mean activation vectors.

Specialization across types. Different typed modules in CAUSALCOMP produce near-orthogonal activations (mean cosine 0.066), in stark contrast to within-type seen/unseen similarity (0.998). This 15× ratio is the structural fingerprint of mechanism learning: each module operates in its own representational subspace.

Table 4: Capacity-matched ablation on SimplePhysics (single seed; absolute values differ from Tables 1–2 due to seed variance, but the *opposing trends* within this table are the key finding). Increasing capacity *worsens* SingleModule's gap (+30pp) but *improves* CausalComp's gap (-7pp), ruling out implicit regularization.

Model	Params	Seen MSE	Unseen MSE	Comp Gap ↓
SingleModule-Std (d=128)	316K	**0.030**	**0.058**	96.2%
SingleModule-Big (d=256)	1.25M	0.023	0.051	125.6%
CAUSALCOMP-Std (d=128, M=8)	845K	0.046	0.073	59.6%
CAUSALCOMP-Big (d=256, M=8)	**3.36M**	0.048	0.074	**52.9%**

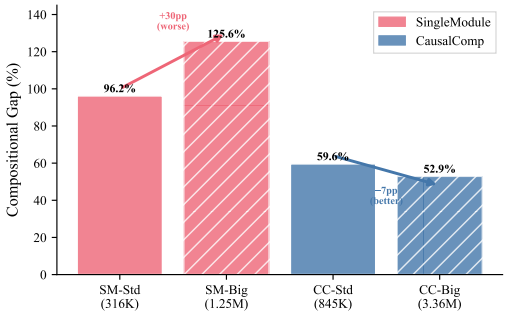

Figure 5: **Capacity ablation.** More capacity worsens SingleModule's gap (+30pp) but improves CausalComp's (-7pp).

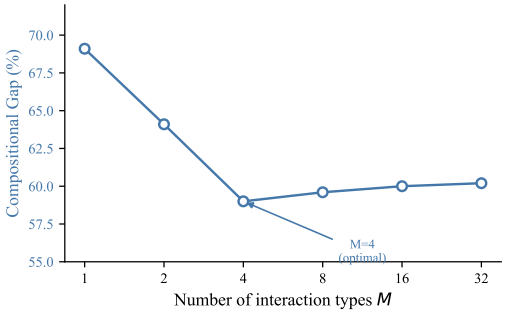

Figure 6: **Sensitivity to M.** Gap follows a U-shape: M=4 is optimal, with mild increase for M>8.

Why specialization matters more than transferability. A natural alternative hypothesis is that any well-trained module shows high seen/unseen activation similarity (i.e., transferability is generic, not specific to typed modules). We confirm this: SingleModule's seen/unseen cosine is 0.975, nearly as high as CAUSALCOMP's 0.998. Activation-level generalization alone does *not* distinguish memorization from mechanism learning. What does distinguish them is whether the architecture forces interactions to be routed through specialized, non-overlapping modules. CAUSALCOMP does; SingleModule cannot. This explains why SingleModule achieves low absolute MSE (its single module generalizes well at the activation level) but high compositional gap (it conflates distinct interaction mechanisms into one representational subspace).

4.6 Sensitivity to M

We vary the number of interaction types $M \in \{1, 2, 4, 8, 16, 32\}$ on SimplePhysics. The compositional gap follows a U-shape: decreasing from 69.1% (M=1) to a minimum of 59.0% (M=4), then mildly increasing to 60.2% (M=32). Too few types conflate distinct mechanisms; too many fragment the training data per module. The sweet spot (M=4–8) balances expressiveness with sufficient training signal per type.

4.7 Autoregressive rollout analysis

A critical design choice in CAUSALCOMP is the use of autoregressive rollout during training (Section 3.6). Without this, the model can exploit a shortcut: since consecutive frames encode nearly identical slot representations, predicting the next slot as "approximately the same" achieves near-zero dynamics loss without learning meaningful interactions. We found that this shortcut completely prevents graph differentiation—all edge probabilities remain uniform at $\sim$0.14 (= K/K^2 where K=8 is the number of slots).

Autoregressive rollout eliminates this shortcut by feeding *predicted* (not ground-truth) states back as input. Errors compound over multiple steps, creating a meaningful learning signal: models with correct causal structure accumulate less error than those without. In our experiments, the dynamics

Table 5: Representation analysis. Each typed module activates nearly identically on seen and unseen object pairs (cosine 0.998), while different types are near-orthogonal (cosine 0.066). This $15\times$ ratio confirms typed modules learn specialized, non-overlapping mechanisms.

Comparison	Cosine Similarity
CAUSALCOMP: same type τ, seen vs. unseen	$\mathbf{0.998 \pm 0.001}$
CAUSALCOMP: different types $\tau \neq \tau'$ (averaged)	0.066 ± 0.091
SingleModule: seen vs. unseen	0.975

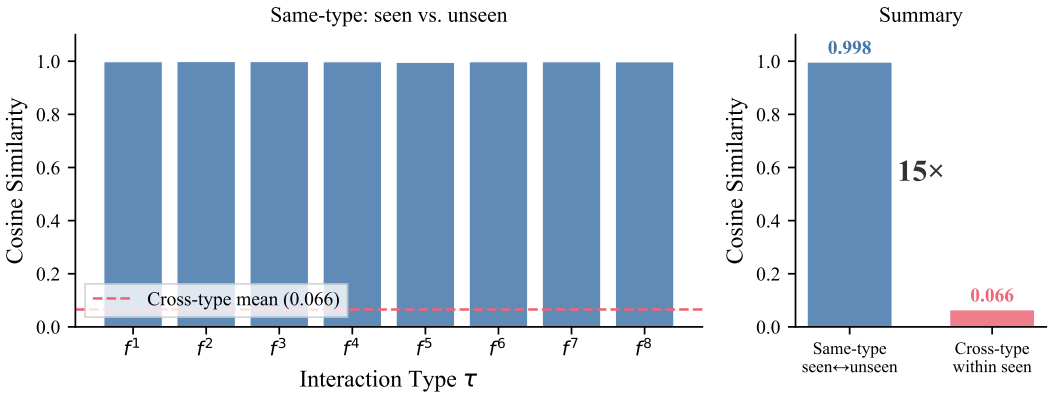

Figure 7: **Representation specialization.** Left: per-type cosine similarity between seen and unseen activations (all $>$0.995). Red dashed line shows cross-type mean (0.066). Right: the $15\times$ gap between same-type and cross-type similarity confirms typed modules learn specialized, non-overlapping mechanisms.

prediction loss stabilized at 0.085 with autoregressive rollout (vs. 0.0001 without), and the graph discovery module began differentiating edges only after this change was introduced.

5 Discussion and limitations

Implicit regularization is ruled out. Section 4.4 and Table 4 directly test whether typed modules' advantage is merely capacity limitation. The 2×2 ablation shows that increasing capacity makes SingleModule's gap *worse* ($+$30pp) while making CAUSALCOMP's gap *better* ($-$7pp)—even when CAUSALCOMP has $2.7\times$ more parameters. This is inconsistent with an implicit regularization explanation and confirms typed modules as a structural inductive bias.

When does CAUSALCOMP fail? CAUSALCOMP struggles when (a) the environment has only one interaction type (NRI Springs), where typed modules provide no advantage; (b) edge supervision does not cover all interaction types—on MultiPhysics, supervision uses only collision events, missing continuous force-based interactions (gravity, charge). Extending supervision to proximity-based proxies for continuous forces is a natural next step that we expect to further improve unseen MSE; and (c) the number of types M is misspecified—too few types conflate distinct mechanisms, while too many fragment the training data per module (Section 4.6).

Limitations. (1) Our environments are synthetic 2D physics with GT object states. Extending to visual observations (e.g., via Slot Attention or DINOv2) and 3D environments (e.g., Physion [Bear et al., 2021]) is critical future work. (2) M is a hyperparameter. Automatic model selection for the number of interaction types (e.g., via a Dirichlet process prior) would improve applicability. (3) Edge supervision uses GT collision events. Unsupervised causal discovery remains open. (4) We do not evaluate on downstream tasks (planning, model-based RL). Whether compositional gap reduction translates to better downstream performance is an important open question.

6 Conclusion

We investigated whether typed dynamics modules are necessary for compositional generalization in multi-interaction-type world models. Through controlled ablations across three physics environments, we found that causal graph discovery alone is insufficient: a model with learned graph structure but shared dynamics (SingleModule) achieves low absolute error but the worst compositional gap (up to 75% on SimplePhysics, 126% when capacity is increased), while CAUSALCOMP's typed modules reduce the gap by 31–34%. Our representation analysis (Section 4.5) further reveals that SingleModule's failure is not from poor representation transfer *per se*—both architectures generalize at the activation level (cosine >0.97). Rather, the absence of *specialization* across interaction types is what enables pair-specific memorization to surface in prediction error. This establishes typed modules as a necessary structural inductive bias: not because they improve transfer, but because they enforce non-overlapping mechanism representations. Important limitations remain—our environments are synthetic with GT states, and the gap reduction does not translate to lower absolute MSE than high-capacity baselines. Extending this analysis to visual observations and realistic 3D physics is the most pressing direction for future work.

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A Implementation details

Architecture. State encoder and decoder are 2-layer MLPs with ReLU activation. Edge MLP and type MLP are 3-layer MLPs. Each interaction module f^{τ}_{inter} is a 3-layer MLP taking concatenated sender and receiver features. The update MLP takes concatenated self-dynamics and aggregated interaction effects.

Training. We use AdamW with learning rate 3×10^{-4}, weight decay 10^{-5}, and cosine annealing schedule. Gradient clipping at norm 1.0. Phase 1 (reconstruction only): 10 epochs. Phase 2 (full training): remaining epochs. Autoregressive rollout of 8 steps during training.

Synthetic environment. Objects are colored circles with radius $\in \{4, 6, 8\}$ pixels on a 64×64 canvas. Velocities sampled uniformly from $[-3, 3]$ pixels/frame. Elastic collisions with equal mass. Wall bouncing with perfect reflection.

B Qualitative results

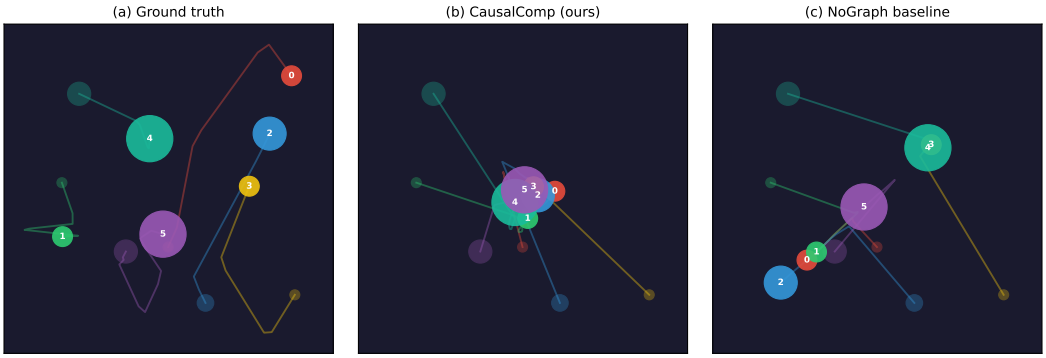

Figure 8: **Trajectory prediction comparison.** (a) Ground-truth trajectories of 6 objects over 16 frames. (b) CAUSALCOMP predictions closely track ground truth, including after collisions. (c) NoGraph baseline shows drift due to inability to model inter-object interactions. Filled circles show final positions; trails show full trajectories.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract claims that typed interaction modules are necessary for compositional generalization (supported by SingleModule vs. CausalComp ablation in Tables 1–2, capacity analysis in Table 4, and representation transferability in Table 5). The abstract explicitly states we do not claim state-of-the-art absolute prediction accuracy.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 discusses four specific limitations: (1) synthetic-only environments, (2) number of interaction types as a hyperparameter, (3) edge supervision requiring ground-truth collision events, and (4) GT object states without end-to-end validation.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper provides theoretical motivation (Section 3.5) based on the Independent Causal Mechanisms principle and causal factorization, but does not claim formal theoretical results. The arguments are informal and clearly stated as such.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Appendix A provides full architecture details (layer sizes, activation functions), training details (optimizer, learning rate, schedule, number of epochs), and environment specifications. All synthetic environments are procedurally generated from described parameters.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code will be released as supplementary material. All datasets are procedurally generated (no external data downloads required). The NRI Springs benchmark follows the standard protocol from Kipf et al. (2018).

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 4.1 describes environments, compositional splits, and metrics. Section 4.4 reports the hyperparameter sweep. Appendix A provides full training details including optimizer (AdamW), learning rate (3×10^{-4}), weight decay, gradient clipping, and number of epochs (150).

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All results in Tables 1–3 report mean $\pm$ standard deviation over 3 random seeds. The seeds control both data generation and model initialization. Error bars capture variability from random data splits and parameter initialization.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Experiments were conducted on a single NVIDIA A100 80GB GPU. Each model trains in approximately 30–90 minutes depending on environment complexity. The full evaluation (3 environments $\times$ 5 methods $\times$ 3 seeds) requires approximately 50 GPU-hours total.

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Justification: This work develops a method for physics prediction in synthetic environments. It does not involve human subjects, personal data, or dual-use concerns.

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Justification: This is foundational research on world models for physical dynamics prediction. It does not have direct negative societal impacts. Potential positive impacts include improved robot planning and physical reasoning, but these are distant applications.

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Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The NRI Springs benchmark is cited (Kipf et al., 2018). Our implementation follows the standard protocol. All other environments and code are original.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: We introduce two new synthetic environments (SimplePhysics, MultiPhysics) and compositional evaluation splits. These are fully documented in Section 4.1 and Appendix A, with code provided as supplementary material.

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